
SUMMARY

- Final-stage PhD researcher in Artificial Intelligence and Machine Learning with experience building and deploying end-to-end ML pipelines across healthcare, analytics, and document-rich domains.
- Developed solutions using Python, SQL, and ML frameworks such as scikit-learn, TensorFlow, Keras, and PyTorch; integrated reproducibility, metadata tracking, and performance benchmarking into workflows.
- Worked with pretrained LLMs (GPT, BERT) for summarisation, representation learning, and document-centric NLP tasks using Hugging Face and OpenAI APIs.
- Worked with cloud-based tools (Databricks, Spark) and MLOps frameworks (MLFlow, GitHub Actions, Docker) to ensure scalable, reproducible delivery.
- Experience collaborating with cross-functional stakeholders, including clinicians, software engineers, and analysts, to align modeling solutions with strategic priorities.
- Strong communicator with experience producing stakeholder-facing summaries, visual insights, and technical documentation that support product delivery.

EXPERIENCE

AI/ML Engineer – Pipeline Development & AI Experimentation (PhD) Sep 2021 – Present
University of Limerick Limerick, Ireland

- Developed modular AI/ML pipelines using Python and SQL for healthcare and document-centric datasets; implemented reproducibility, validation, and Git-based version control.
- Applied supervised learning, model tuning, and feature engineering across structured and semi-structured data; contributed to LLM/NLP workflows in classification and summarisation tasks.
- Generated dashboards and performance summaries to communicate model outcomes and diagnostic insights to cross-disciplinary stakeholders.

Applied Researcher – Edge AI & MLOps Foundations (PhD) Jan 2023 – Dec 2023
University of Limerick Limerick, Ireland

- Built real-time analytics pipelines on embedded devices (Nordic Thingy:53) for health monitoring use cases; optimised for latency, logging, and system resource constraints.
- Contributed to MLOps pipelines including metadata capture, model evaluation automation, and modular design for scalable experimentation and traceability.

Academic Teaching & Mentoring Experience Jan 2021 – May 2023
University of Limerick & National College of Ireland Ireland

- Taught and mentored in AI/ML, Data Analytics, and Software Testing; embedded real-world healthcare/NLP projects to highlight ML pipeline design and reproducibility principles.
- Supervised student projects using Python, R, SQL, Tableau and Power BI; focused on predictive modelling, summarisation, and stakeholder-aligned data communication.

PUBLICATIONS & PROJECTS

- **Full Google Scholar Profile for all publications (10+):** Amandeep Singh – [shorturl.at/egmD4](https://scholar.google.com/citations?user=egmD4)
- CINCO de Bio: A Future Internet Platform for Domain-Specific Workflows that Leverage AI for Biomedical Research | PhD AI/ML | doi.org/10.3390/biomedinformatics4030102
- IDPP: Imbalanced Datasets Pipelines in Pyrus (Tool Review) | PhD AI/ML | shorturl.at/lyXZ8
- Model Driven Development for AI-Based Healthcare Systems: A Review | PhD AI/ML | doi.org/10.1007/978-3-031-73741-1_15
- Low-Code Internet of Things Application Development for Edge Analytics | PhD AI/ML | doi.org/10.1007/978-3-031-18872-5_17
- Enhancing Decision-Making for Imbalanced Medical Datasets Using BDDs and Low-Code/No-Code | PhD AI/ML | doi.org/10.1109/MITP.2024.3459248
- CNN-based Human Activity Recognition (HAR) on Edge Computing Devices | PhD AI/ML | doi.org/10.1109/COINS57856.2023.10189270
- **Other Projects:** Transfer learning for image classification, COVID-19 analytics, Power BI for enterprise reporting, predictive modelling using retail and environmental data, masked face detection using CNNs, and more.

EDUCATION

University of Limerick

PhD in Artificial Intelligence and Machine Learning (AI/ML)

Limerick, Ireland

Sep 2021 – May 2025

Fully-funded by Science Foundation Ireland Centre for Research and Training in Artificial Intelligence (SFI CRT-AI).

Modules: Machine Learning, Optimization and Constraint Programming, Visual Media Processing, Reinforcement Learning and Personalisation, Natural Language Processing, Research Integrity, Digital Research Management, Research Networking: Developing an Academic Profile, Planning Research & Publication, Research Ethics, Developing Ideas & Arguments: Writing into Academic Communities.

National College of Ireland

M.Sc. in Data Analytics (DA)

Dublin, Ireland

Jan 2020 – Feb 2021

Grades: 77.1%, 1.1 grade, First Class Honours.

Modules: Data Mining and Machine Learning-I (DMML-1), Database and Analytics Programming, Business Intelligence and Business Analytics, Statistics for Data Analytics, DMML-2, Modelling Simulation and Optimization, Domain Application of Predictive Analytics, and Data Governance and Ethics.

University of Zurich

M.Sc. in Physics - Theoretical Astrophysics & Cosmology

Zurich, Switzerland

Sep 2017 – Oct 2019

Grades: 4.5/6, 2.1 grade.

Modules: Theoretical Astrophysics, Theoretical Cosmology, Astro-Particle Physics-I & II, Introduction to Astrobiology, Extra-Solar Planets, The Sun & the Planets, Planet Formation.

University of Delhi

B.Sc. (Honours) in Physics

New Delhi, India

Jul 2014 – Jul 2017

Grades: 77.64%, 1.1 grade, First Class Honours.

Relevant Modules: Mathematics (Analysis & Statistics), Mathematical Physics (Linear Algebra, Calculus, Linear Programming), Numerical Analysis (C++), Microprocessor Programming, Digital Electronics.

TECHNICAL SKILLS

- **Languages & Frameworks:** Proficient in Python and SQL; experience with Scikit-Learn, TensorFlow, Keras, and Hugging Face for predictive modelling, summarisation, and text representation in medical and document-heavy datasets.
- **Data Pipelines & Engineering:** Built ingestion, transformation, and automation workflows for healthcare and analytics data using Python and low-code tools; integrated pipelines with downstream ML and reporting workflows.
- **MLOps & Deployment:** Familiar with CI/CD workflows, test automation, and reproducibility practices; experience with GitHub Actions, MLFlow, Docker, and FastAPI to support scalable, deployable ML services.
- **Big Data & Cloud Readiness:** Exposure to PySpark and Databricks for distributed ML development; familiar with cloud analytics workflows and data architecture concepts in Snowflake.
- **Data Engineering & Visualization:** Designed transformation pipelines using Python and SQL; delivered insights through dashboarding tools and custom visual summaries.
- **Collaboration & Agile Delivery:** Contributed to Agile teams; worked cross-functionally with engineers, analysts, and stakeholders to align model outcomes with product goals.

INTERESTS & ACHIEVEMENTS

- **Technical Projects:** Developing a personal website (HTML/CSS/JS) and building a custom NAS/server from scratch. Deployed services such as Immich, Jellyfin (*ARR stack), and others using Docker, with focus on automation, container orchestration, and performance tuning.
- **Community:** Volunteer at the local Gurudwara (Sikh Temple) in Dublin, supporting food and clothing distribution initiatives for those in need.
- **Education Outreach:** Organise and deliver free Physics and Mathematics classes for high school students, helping them adapt to digital learning challenges.
- **Recognition:** Awarded the KVS National INSPIRE Award by the Department of Science & Technology, Government of India (2012); Grade 12 School Topper (2014).
- **Music:** Holder of a Diploma in Harmonium (Vocal) from Prayag Sangeet Samiti; active in devotional music performance.